Megha Sinha

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Summary

Software Engineer with 2 years of experience at Oracle Cloud Infrastructure building scalable backend services, distributed subscription workflows, and high-availability REST APIs. Strong track record of cutting production error rates, automating recovery pipelines, and independently driving multi-realm service migrations.

Education

National Institute of Technology, Rourkela **November 2020 – June 2024**
Bachelor of Technology in Computer Science and Engineering *Odisha, India*

Work Experience

Oracle Cloud Infrastructure **July 2024 – August 2026**
Member of Technical Staff *Bangalore*

- Designed and implemented backend APIs and asynchronous workflows for OCI's **subscription lifecycle service** across commercial, government, and sovereign cloud realms, delivering a zero-touch activation architecture for customer onboarding and data synchronization.
- Reduced on-call incident volume by **25%** by engineering self-service order cancellation workflows, automating cross-team ticket routing, and handling edge cases like suppressed customer emails gracefully to prevent silent failures.
- Resolved **5,200+ stuck workflow executions** during a production incident by isolating root-cause bottlenecks and collaborating with downstream partner teams to execute targeted recovery operations.
- Eliminated **30+ daily production 5XX errors** and restored service health by reclassifying unhandled missing-subscription exceptions, eliminating redundant cross-realm request fanout, and tightening input validation.
- Executed the migration of **6 core lifecycle and contract-update workflows** to a new downstream architecture across realms while maintaining full backward compatibility for zero-downtime deployment.
- Eliminated annual SMTP credential rotation toil by centralizing email notifications onto Announcement Service, built an event-driven activation workflow, and developed a host-level backfill tool to reconcile historical subscription data across all OCI realms.

Oracle Cloud Infrastructure **May 2023 – July 2023**
Project Intern *Bangalore*

- Provisioned OCI cloud infrastructure using **Terraform**, optimized database performance by migrating legacy Criteria queries to **HQL**, and established comprehensive **Grafana** monitoring dashboards, alerts, and operational runbooks.

Projects

AgentMock – MCP Contract Testing CLI *[GitHub] | TypeScript, Node.js, MCP, JSON-RPC 2.0, Zod*

- Engineered a CLI tool to record, replay, and detect breaking schema changes for Model Context Protocol (MCP) tool servers, isolating AI agent tests from slow third-party backends.
- Built an in-process mock replay server over `stdio` and JSON-RPC, cutting local test latency from 3-4s cold starts to <4ms with zero external API costs.

Multi-LLM Gateway – Fault-Tolerant AI Routing Proxy *[GitHub] | TypeScript, Bun, Elysia, PostgreSQL, Prisma*

- Built a unified chat API routing requests across OpenAI, Claude, and Gemini with PostgreSQL-backed model mappings, API key validation, and token credit accounting.
- Engineered automatic provider failover and bounded retries with error classification, skipping retries for auth errors while transparently retrying transient failures.

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, C/C++, SQL
Backend & Web: REST APIs, Microservices, Asynchronous Workflows, Elysia, Bun, React
Databases & Persistence: PostgreSQL, MySQL, Prisma, HQL
Cloud & DevOps: Oracle Cloud Infrastructure (OCI), Terraform, Docker, CI/CD, Deployment Automation
Reliability & Observability: Grafana, Monitoring & Alerting, Retry Handling, Rate Limiting
Tools & Testing: Git, Maven, Postman, Integration Testing

Achievements

Oracle Platform Spot Bonus Award: Recognized for delivering key features and reliability improvements in the subscription lifecycle platform across multiple OCI environments.